# Krimisizer

Appropriate sound effects are often used to provide background audio for radio dramas and disco shows; the siren circuit described here generates just such an effect. Almost everyone is familiar with the difference between the "wee-woo" sound of European police sirens and the nervous wail that heralds the arrival of their American counterparts. The "Krimisizer" is capable of producing both sounds.

The block diagram in Figure 1 illustrates the circuit's operating principle. An astable multivibrator (AMV) generates a voltage; for the American siren mode, this voltage is converted into a sine-like signal using an integrator and a low-pass filter. This signal controls a voltage-controlled oscillator (VCO). For the European siren mode, the square-wave signal is fed directly to the VCO, bypassing the integrator.

The active components in the astable multivibrator (AMV) are transistors T1 and T2 (see Figure 2). Different clock frequencies are required for the two types of siren. For the "E" siren (switch in position "E"), components R2, R3, P1, and C2 determine the frequency; for the "A" siren, R3, P2, and C2 determine the "wail" rate. Diode D1 protects transistor T1 against excessive base-emitter reverse voltage. Potentiometer P3 adjusts the amplitude of the integrated signal, which determines the "wail frequency" of the "A" siren. Along with P3, the values of R10 and C5 also influence the integrator amplitude. The R11/C4 network converts the original triangular voltage into a signal with a sine-like waveform. This voltage controls the VCO frequency to produce the characteristic siren sound. The siren's fundamental frequency is set using potentiometer P6. Strictly speaking, however, it is incorrect to call it a VCO; it is actually a current-controlled oscillator.

In switch position E, the circuit generates the European siren tone. The square-wave voltage tapped from potentiometer P4 causes the VCO to switch abruptly between two fixed frequencies. Potentiometer P5 is used to set the lower tone frequency, while P4 adjusts the "step height"—and thus the frequency of the higher tone. A single 3-pole changeover switch (3xCO) can be used for S1–S3; however, the range of siren sounds is expanded if three independent single-pole changeover switches (1xCO) are used instead.

This module is certainly an addition that leaves synthesizer purists cold, since siren sounds can also be generated in the FORMANT using LFOs. However, the Krimisizer module offers the ability to quickly and easily create typical (police) sirens without a tangle of patch cables—an advantage that is particularly useful for live performance.

# Front Panel and PCB

Practical implementation requires a front panel compatible with the FORMANT system, as well as a printed circuit board (PCB). Figures 4 and 6 show suggested designs.

To use the unit within the FORMANT system, the power supply lines and the output connection must be adapted. The additional circuitry required is shown in Figure 5.

## Assembly Notes

The original circuit design includes a total of six controls (potentiometers or trimmers) and three switches (each a single-pole changeover type). However, since pairs of controls and a switch are functionally linked—and given the limited space available on a standard FORMANT front panel—the capabilities of the KRIMISIZER have been slightly reduced in favor of significantly simplified operation.

A 470k linear dual-gang potentiometer is to be used for P1 and P2, and a 4.7k linear dual-gang potentiometer for P5 and P6. For P3 and P4, it is best to use a dual (tandem) potentiometer with 22k/100k values. After opening the potentiometer housing, one of the resistive tracks must be replaced with a track from a single potentiometer, as dual potentiometers with two different track values are generally not commercially available. If replacing the resistive tracks is not feasible, a compromise solution is to use a dual potentiometer with 47k (linear) values. Switch S1 is associated with potentiometer P1/P2, S2 with P3/P4, and S3 with P5/P6. The switch setting "EI" (European) corresponds to LOW, while "A" (American) corresponds to HIGH.

In practice, it has proven advantageous to set R2 to 470Ω and R10 and R15 to 4.7kΩ.

## Applications

The KRIMISIZER can be fed into the signal path via the CV inputs of the VCF and VCA or used for modulation purposes.

## Parts list for Figure 5

### Resistors:

R20 = 33 k
R21 = 68 k
R22 = 100 k
R23 = 1 k
R24 = 150

### Capacitors:

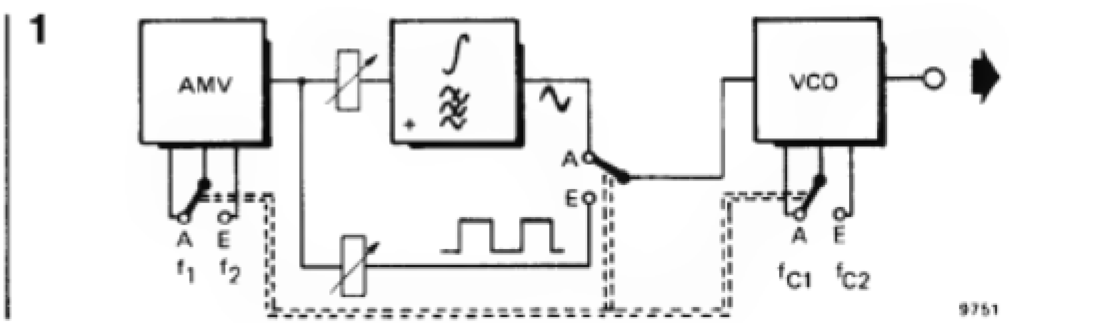
C9, C10 = 100 n

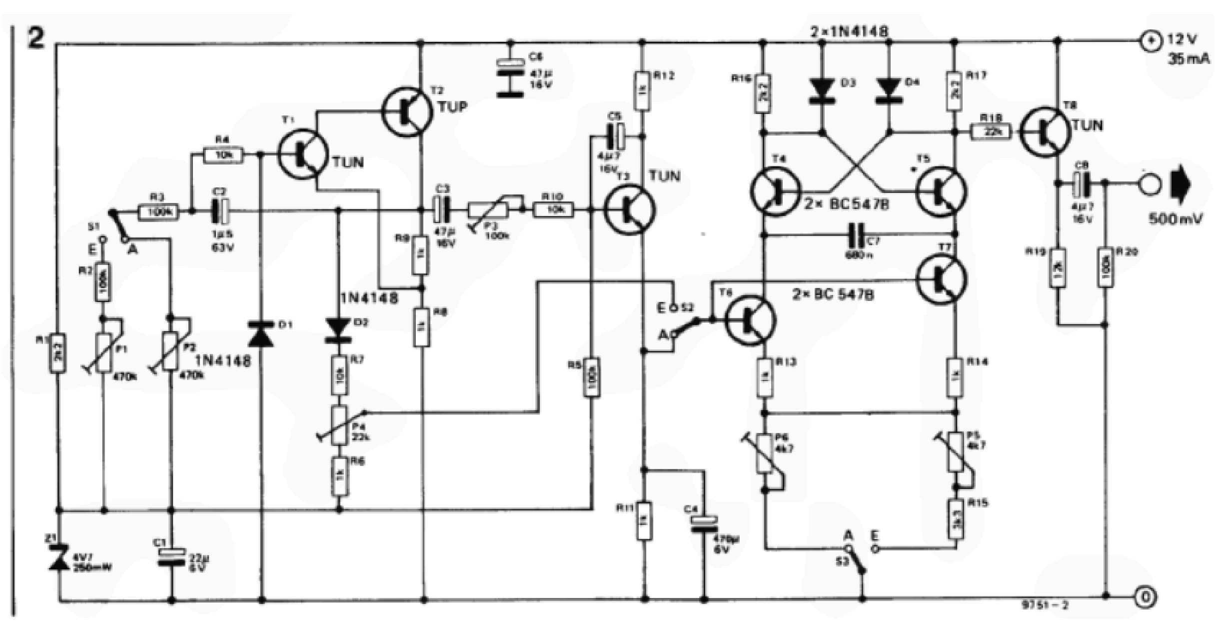
### Semiconductors:

D5 = ZPD 12
IC1 = µA 741C

## Figures

- Figure 1. Block diagram of the siren.
- Figure 2. The circuit. A 3-pole changeover switch can be used for S2–S3; however, using independent switches increases the number of selectable siren sounds.
- Figure 3. PCB and component layout for the "Krimisizer".
- Figure 4. Suggested front panel design for the Krimisizer.
- Figure 5. Adaptation circuit for the Krimisizer within the FORMANT system.
- Figure 6. PCB layout and component placement for the Krimisizer circuit. The board includes the components for the adaptation circuit shown in Figure 5.



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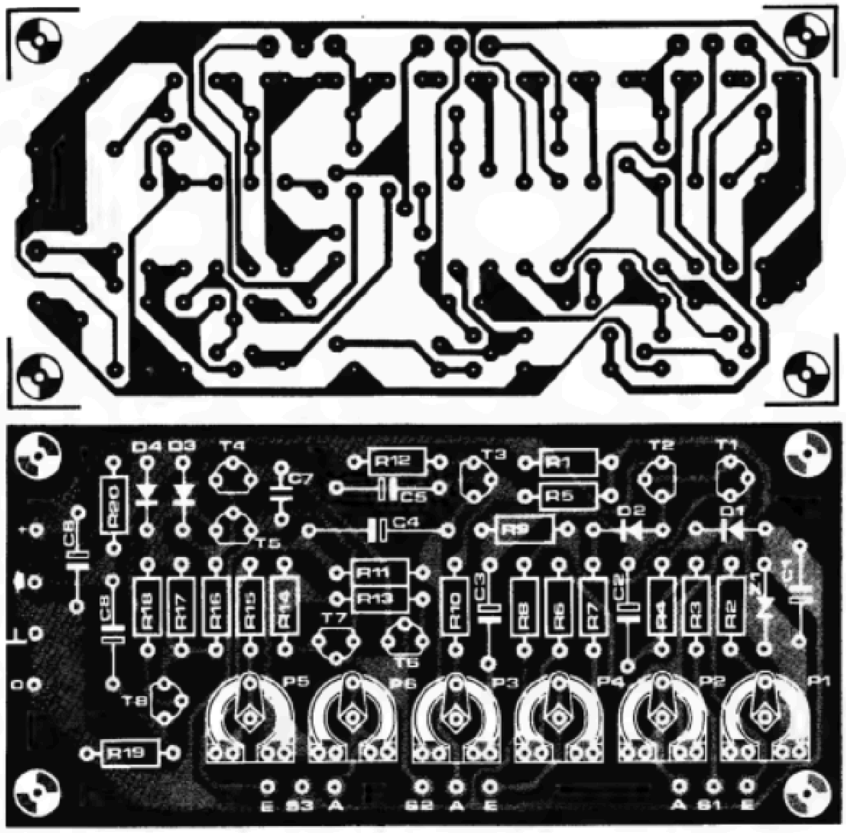
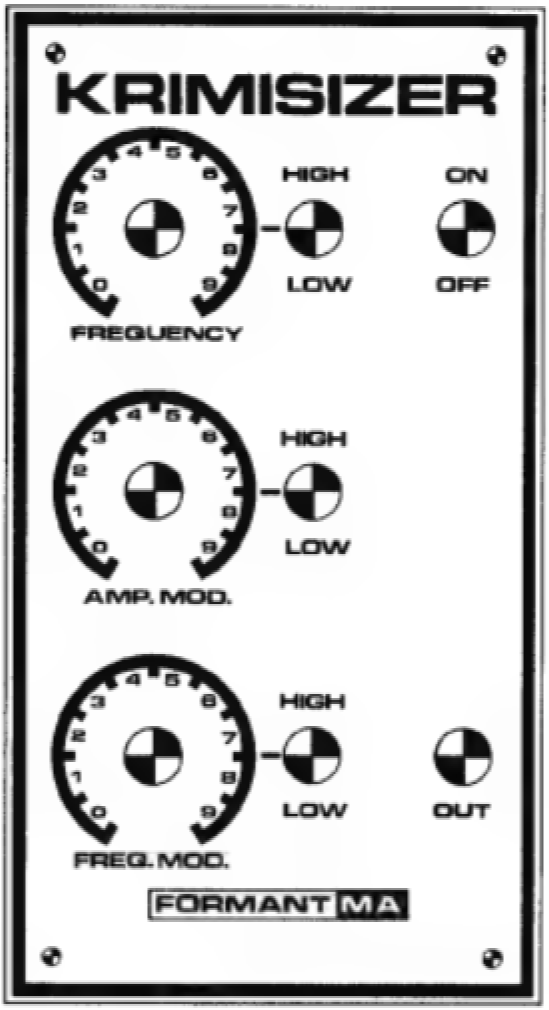
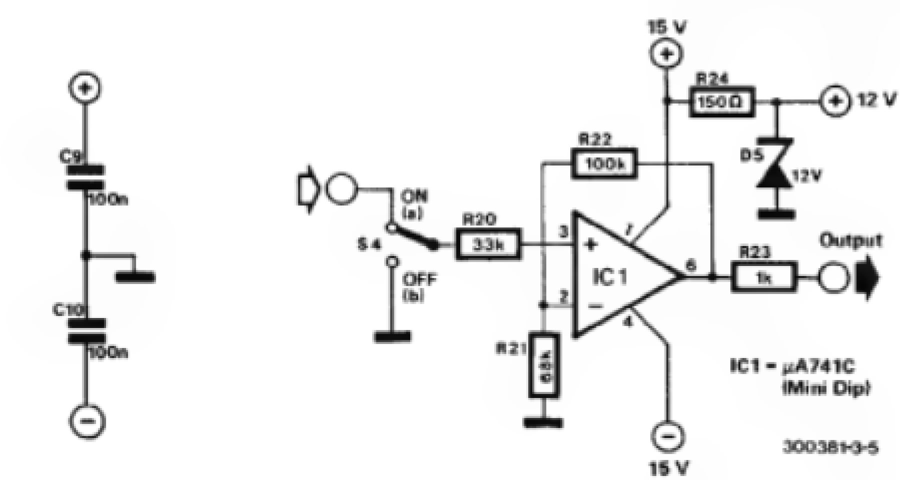

D4 D3 T4 T5 C7 R12 C5 C4 T3 R1 R5 R9 D2 T2 T1 D1
R20 C9 C8 R18 R17 R16 R15 R14 R11 R13 T7 T6 R10 C3 R8 R6 R7 C2 R4 R3 R2 C1
T8 P5 P6 P3 P4 P2 P1 R19
E S3 A S2 A E A S1 E

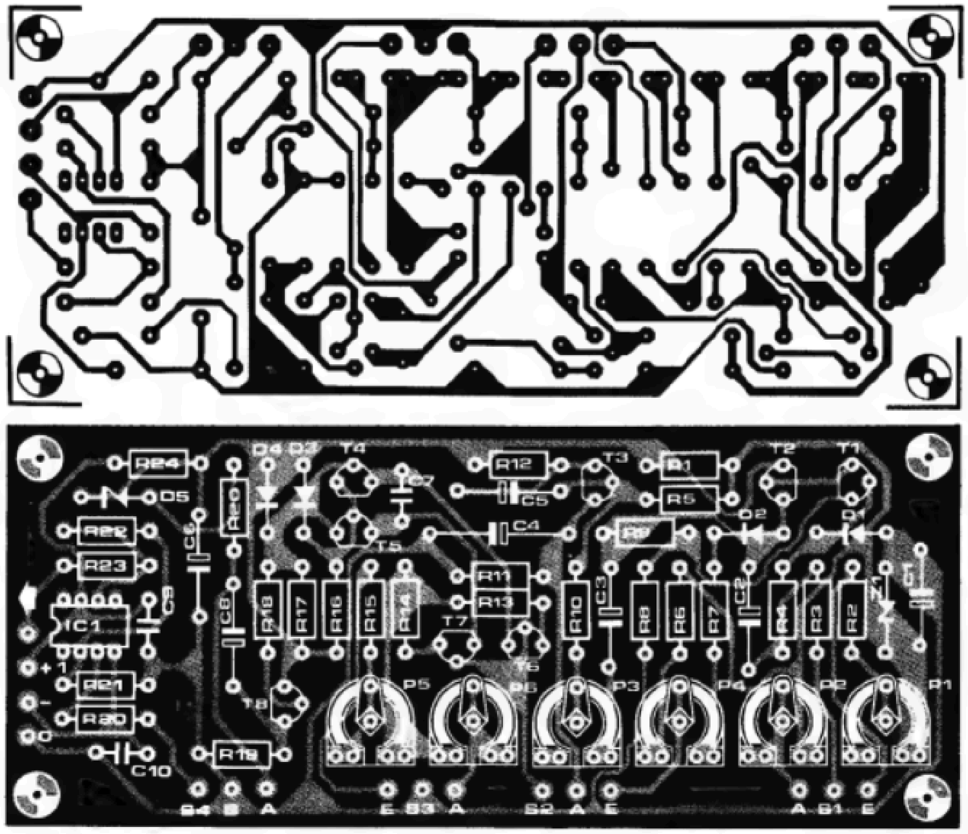
4


KRIMISIZER
HIGH
ON
LOW
OFF
FREQUENCY
HIGH
LOW
AMP. MOD.
HIGH
LOW
OUT
FREQ. MOD.
FORMANT MA

5


C9
100n
C10
100n
15 V
R24
150 Ω
12 V
R22
100k
D5
12V
ON
(a)
S4
OFF
(b)
R20
33k
IC1
R21
68k
R23
1k
Output
15 V
IC1 = µA741C
(Mini Dip)
300381-3-5

6



R24
D5
R22
R23
IC1
R21
R20
C10
R19
D4
D3
T4
C7
T5
R12
C5
C4
T3
R1
R5
R9
D2
D1
T2
T1
C6
R20
C9
C8
R18
R17
R16
R15
R14
R11
R13
T7
T6
R10
C3
R8
R6
R7
C2
R4
R3
R2
Z1
C1
T8
P5
P6
P3
P4
P2
P1
+
-
0
S4
B
A
E
S3
A
S2
A
E
A
S1
E